



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT – PART II

In addition to part I (General Handout for all courses appended to the Time table), this portion gives further specific details regarding the course.

Course No : BITS F471
Course Title : Introduction to Large Language Models
Instructor-in-Charge : Dr. Dhruv Kumar (dhruv.kumar@pilani.bits-pilani.ac.in)

1. Course Description:

Introduction to Large Language Models (LLMs); prompt engineering techniques (e.g., zero-shot, few-shot, chain-of-thought), autonomous prompt optimization; retrieval-augmented generation (RAG), vector databases, multi-modal RAG methods; tool and function calling; multi-agent workflows (memory, planning, action, reflection); safety, guardrails, reliability; LLM-based evaluation, human evaluation; life cycle of LLM training, LLM architecture, fine-tuning LLMs using supervised learning or reinforcement learning; inference optimization; Mechanistic Interpretability; real-world case-studies

2. Scope and Objectives:

By the end of this course, students will be able to:

- Understand and apply prompt engineering techniques to control the behavior of LLMs.
- Build retrieval-augmented generation (RAG) pipelines for combining LLMs with external knowledge bases.
- Fine-tune LLMs for specific tasks using publicly available models.
- Leverage tool calling to enable LLMs to interact with external systems and APIs.
- Design and implement multi-agent workflows where multiple LLMs interact to solve complex tasks.
- Implement guardrails to ensure the safe and ethical deployment of LLMs in production environments.
- Understand the internal architecture of LLMs and the entire life cycle of LLM Training and Fine-Tuning.
- Evaluate the performance and accuracy of LLM-powered applications using appropriate metrics.
- Understand the principles of inference for deploying LLM models efficiently at scale.

3. Course Plan:

Lectures	Topic	Learning Outcomes	References
1-3	Prerequisites [Basics of Machine Learning and Natural Language Processing]	High level overview of training and inference process in machine learning and natural language processing [popular tasks such as regression, classification, next token prediction etc.], Understand importance of data quantity and quality	Class Notes
4-5	Introduction to LLMs	Understand LLM foundations, Identify LLM use cases, Understand LLM architectures at a very high level, Understand API access to LLMs	T1, T3
6-7	Prompt Engineering	Understand and apply various prompt techniques such as Zero-shot prompting, Few-shot prompting, Chain-of-Thought (CoT), Role Playing, Self-consistency, Autonomous Prompt Optimization	T1, T3



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Lectures	Topic	Learning Outcomes	References
8-9	Retrieval-Augmented Generation (RAG) and Vector Databases	Understand RAG principles, Build RAG pipelines, Assess trust in RAG systems, Understand vector databases, Implement similarity search, Multi-Modal RAG	T1
10-11	Tool Calling	Implement tool calling with LLMs, Integrate APIs into workflows, Build multi-tool agents, Use API calls for real-world tasks	T1
12-14	Multi-Agent Workflows	Understand multi-agent systems, Implement memory-driven agents, Plan with agent-based workflows, Use reflection for decision-making	T1
15-16	Guardrails for Safe LLM Deployment	Understand LLM risks, Apply safety guardrails, Mitigate hallucinations, Red Teaming, Blue Teaming	T1
17-18	Evaluation of LLM-Powered Applications	Understand key evaluation metrics, Compare human and automated methods, Evaluate task-specific LLM applications	T1, T2, T3
19-21	Model Training	Data Preparation, Tokenization, Embeddings, Pre-Training, Fine Tuning, Human Feedback	T1, T2, T3
22-25	LLM architecture	Attention, Multi-Head Attention, Transformers, GPT Architecture	T3
26-28	LLM Inference	, Temperature, Top-K Sampling, Top-P Sampling, Test Time Compute	T1, T2, T3
29-31	Fine-Tuning Large Language Models	Understand fine-tuning methods, Implement LoRA for efficiency, Use Supervised fine-tuning (SFT) for specialized tasks, Apply fine-tuning in low-resource settings	T1, T2, T3
32-36	Reasoning LLMs and Reinforcement Learning for LLM Fine-Tuning	Eliciting CoT Reasoning in LLMs, Reinforcement Fine-Tuning, Proximal Policy Optimization (PPO), Direct Preference Optimization (DPO), Group Relative Policy Optimization (GRPO)	T2
37-39	Inference Optimization and Scaling LLMs in Production	Understand efficient LLM inference, Understand KV cache management, Optimize memory and compute costs, Implement batching and parallelism with vLLM, Apply scaling strategies in production	T1
40-42	Mechanistic Interpretability	Activation Patching, Activation Steering, Linear Probing, LogitLens, Circuits	Class Notes

4. Textbooks:

The field of LLM Engineering is rapidly evolving and still in its early stages. Hence, in addition to using textbooks, we will also be directly covering the contents of this course from research papers published in top-tier CORE A* conferences and journals such as ICLR, NeurIPS, NAACL and ACL. The textbooks are listed below:

T1 [AI Engineering by Chip Huyen Publisher\(s\): O'Reilly Media, Inc. ISBN: 9781098166304](#)



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T2	Reinforcement Learning from Human Feedback by Nathan Lambert, Online.
T3	Raschka, Sebastian. Build A Large Language Model (From Scratch). Manning, 2024. ISBN: 978-1633437166.

5. Evaluation Scheme

Component	Weightage	Date	Mode
Comprehensive Exam (180 minutes)	35%	Refer to time-table	Partly Open Book
Mid-semester Evaluation for Project	25%	Week Before mid-sem	Take Home / In-person Viva
End-semester Evaluation for Project	25%	Week before comprehensive exam	Take Home / In-person Viva
Final Evaluation for Project	10%	Last day of comprehensive exams	Take Home / In-person Viva
Labs + Guest Lectures + Misc	5%	Weekly / Bi-Weekly	Partly Open Book

- 6. Chamber Consultation:** Book one day before via Email
- 7. Notices:** All the notices will be displayed on the course page at Nalanda-LMS
- 8. Make-up Policy:** Make-up will be granted only in genuine cases based on individual's need and circumstances.
- 9. Academic Honesty and Integrity Policy:** Academic honesty and integrity is to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.